SUBJECT: QBA 1 DATE: / /

quantitative business analysis

* quantitative analysis: scientific approach → managerial, to make better decisions. Data → processing → information.
  - Computers, excel.
  - mathematical tools → solving problems.

* quantitative
  → How much Inv?

* qualitative
  → weather
  → rules, regulations.

* Business analytics: Data driven approach that allows companies to make better decisions.

* descriptive
  - what's happened?
  - what's happening?
  - Consolidation of historical data that allows us to understand how the company performance of the past and now.

* predictive
  - Forecasting the future out comes.

* prescriptive
  - optimization model methods

Steps of OA

1. defining the problem:
   - * clear statement
     - → small No. of problems.
   - * poor statement → poor results.

2. developing a model:
   mathematical representation of the problem
   - → variable: quantified measure that can be
     - Changed or vary (unknown)
       - Controlable
       - non Controlable
     - parameter: ... (known) inherent in the model

3. Acquiring input data:
   garbage in → garbage out.
   improper data → misleading data

4. developing a solution:
   1- Solving the equation
   2- trial & error: try various approach until reaching the best result
   3- Coplete enormation: trying all possible values for the variables in the model to arrive the best solution.
   4- algorithm: repeting few simple steps until finding the best solution.
      (19th century)

5. Testing the solution:
   testing data and model
   How? ex. interview, other methods (Compare)

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[6] Analyzing the results & senstivity analysis: « post optimality »

- model → approximation of reality.
  - ↳ sensetive the results → Change in model or data.

[7] implementation:

↳ 2 main problems: — المدير معترض
— الحل غير قابل التنفيذ

Section (1)

1- A	2- B	3- B	4- C	5- C
6- d	7- a	8- b	9- a	10- b
11- a	12- b	13- b	14- a	

1- ✓	2- ✓	3- ✓	4- X	5- X
6- X	7- ✓	8- ✓	9- ✓	10- X